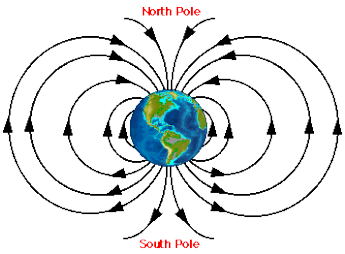
Chapter 45, Magnetism
Calculus Physics II

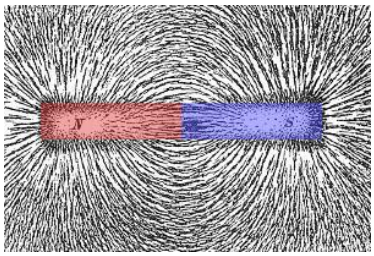
The electron and proton have an electrical charge ($q = 1.6x10^{-19}$ C), this electrical charge produces an electric field, this field exerts forces on charged objects. The electron and proton also have a magnetic moment that produces a magnetic field, this magnetic field interacts with moving charge and material with magnetic properties. The electric charge is the source of the electric and magnetic fields we experience in everyday life.

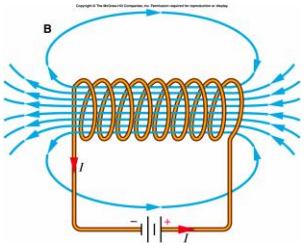
- ✓ Although the magnetic fields and electric fields originate from the same source (charge) they have very different properties!

Magnetic Fields ($\vec{B}$) – A magnetic field exists in an otherwise empty space if a charged object moving through that region experience a force due to its motion in the magnetic field or a magnetic material in that region of space experience a force due to its magnetism.

Source: **Earth** **Permanent Magnetic** **Current in wire**







- ➢ The magnetic field and force are vectors therefore their magnitude and direction must be taken into account.

Units of Magnetic Field ($\vec{B}$)

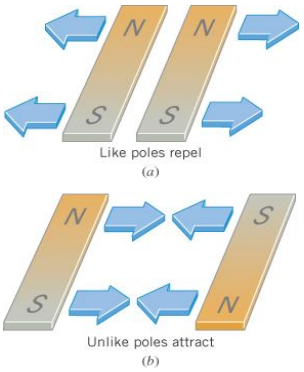
1 Tesla (T) = 1 N/Am = 1 Weber/m^2 or 1×10^{-4} T = 1 Gauss (G)

The Magnetic Field is analogous to other Fields – A field in physics is a condition in space that creates a force when an object is placed in that field with a property that interacts with that field. Charged objects that are moving or material with magnetic properties interact with the magnetic field.

Mass is the property for gravity, charge is the property for electric fields and moving charge is the property for the magnetic field.

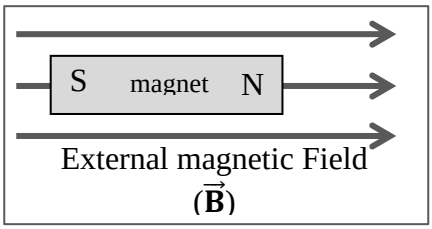
Gravitational Field	–	Force per unit mass [N/kg] $g = F_g/m$
Electric field	–	Force per unit charge [N/C] $E = F_E/q$
Magnetic Field	–	Force per unit moving charge [N/(Cm/s)] $B = F_B/qv$

Magnetic poles - A magnet has two poles which are defined as North and South.

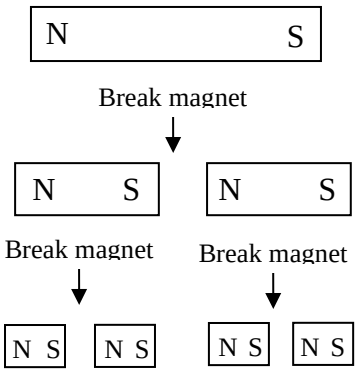


- Poles of the same type repel (N N or S S)
- Unlike poles attract. (N S or S N)

A magnet will align in an external magnetic field with its North Pole in the direction of the external magnetic field and this defines the direction of the magnetic Field $\vec{B}$.



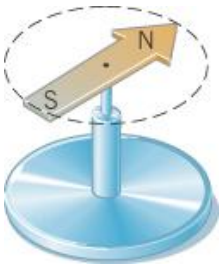
- If a magnet is broken, then the broken pieces form a new magnet with North and South poles for each separate piece this happens all the way down to the atomic scale.
- The source of the magnetism is distributed throughout the magnet not concentrated at the poles or the surface like the electric charge.
- Magnetism is not like static charge and is not free to move around the magnet or to other objects, it is a property of the substance down to the atomic scale.

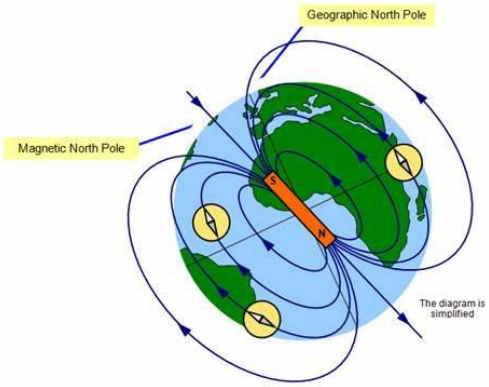


Monopole – a magnetic monopole in physics is a theoretical particle that only has one pole either North or South. Monopoles have never been found in nature!

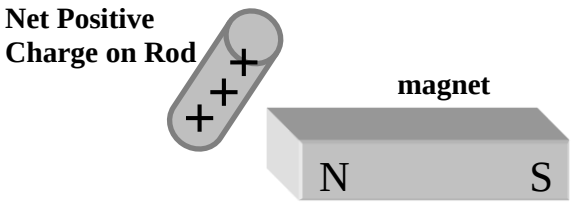
Compass - A compass is a permanent magnet with a north and south pole (magnetic North Pole usually is painted red) that is free to move, balanced on a pivot point. It is used to determine the direction of the magnetic field at or near the surface of the earth.

- A magnetic field is detected by its effect on a compass needle (magnet). The compass needle (north magnetic pole) lines up in the direction of the magnetic field. The compass points long the direction of the magnetic field lines.





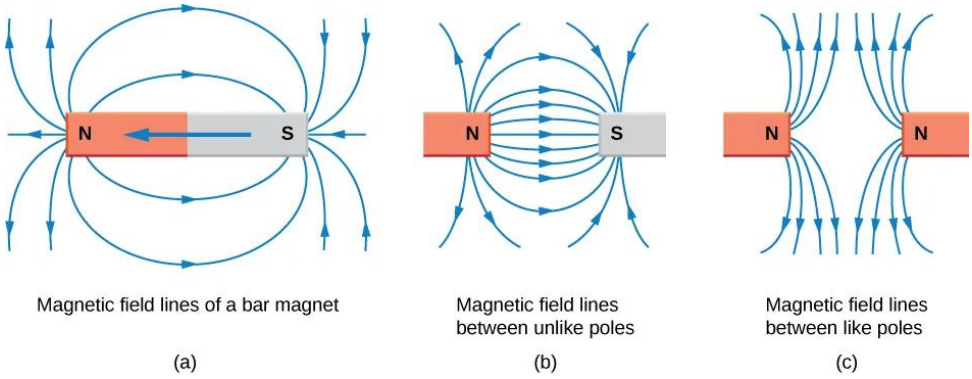
Static Charge and the magnetic field – A charged rod will have no effect on a magnet and the magnetic will have no effect on the charged rod if they are not moving relative to each other.



- If a positively charged rod is brought near a magnet there will be no effect on either, there will be no force on either.
- The Magnetic field is produced by moving charge not by static charge therefore only moving charges are affected by the magnetic fields.
- A static Electric field does not interact with a static magnetic field.

Magnetic Field Lines – The **external** magnetic field from a magnet has Magnetic field lines that exit the North Pole and enter the South Pole and are continuous. Inside the magnet the field lines enter the South Pole and exit the North Pole.

- The direction of the lines is defined by a "test compass" and point in the direction of the magnetic north on the compass.
- Magnetic field lines are closed loop, they are connected throughout their path even through the magnet itself.
- The more lines in an area the stronger the magnetic field.



Magnetic field lines of a bar magnet (a)

Magnetic field lines between unlike poles (b)

Magnetic field lines between like poles (c)

Three Main producers of Magnetic Fields

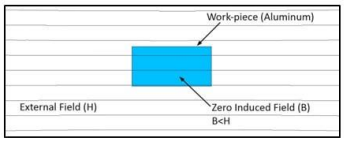
1) Magnetic Material (Loadstone, permanent magnets, Earth)
2) Moving charge (Current carrying wire)
3) The electron (electron spin and electrons moving around the nucleus of the atom) and proton.

The source of magnetism for all magnetic fields is the movement of charge.

- ❖ Magnetic Fields are produced in current carrying wire due to the movement of electrons in the wire.
- ❖ Magnetic fields are produced in Magnets due to the spin and movement of electrons and protons, the alignment of these atoms to produce a net magnetic field.

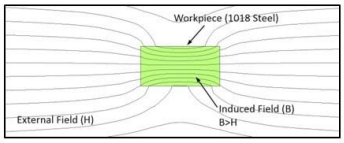
Magnetic Material - The magnetic field produced by a current or magnetic is altered by the presence of any substance. For no substance or vacuum the Magnetic Permeability is μ_o – Permeability of Free Space, $\mu_o = 1.257 \times 10^{-6}$ Tm/A

Applied Field H
Permeability $\mu = 1$
Induced Field B = 0
B < H



Magnetic Permeability (μ) is the amount that material is able to be magnetized, if it is easily magnetized then it has a high permeability.

Applied Field H
Permeability $\mu > 1$
Induced Field $B = \mu \times H$
B > H



Permeability of material (μ) compared to air/vacuum (μ_o)

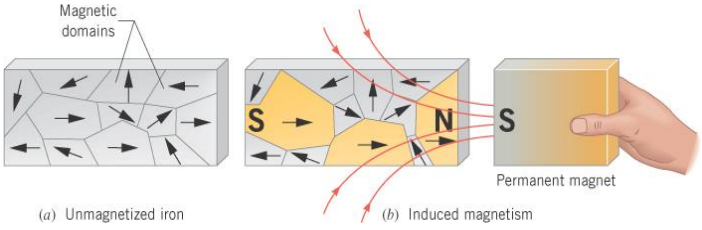
Diamagnetic Material ($\mu < \mu_o$) – A substance that decrease the external magnetic field, this material is not magnetic. Examples: Lead, glass or rubber

Paramagnetic Materials ($\mu > \mu_o$)– A substance that increases the magnetic field by a small amount. Example: Aluminum

Ferromagnetic Materials ($\mu >> \mu_o$)– A substance that greatly increases the magnetic field (100's to 100,000's times). Example: Iron, nickel, cobalt

Material	Permeability (μ) (Tm/A)	Relative Permeability (μ/μ_o)	
Vacuum	1.26×10^{-6}	1	
Air	1.26×10^{-6}	1.0000000037	Paramagnetic
Aluminum	1.26×10^{-6}	1.000022	Paramagnetic
Carbon Steel	1.26×10^{-4}	100	**Ferromagnetic**
Copper	1.26×10^{-6}	0.9999994	Diamagnetic
Pure Iron	**2.5×10^{-1}**	**200,000**	**Ferromagnetic**
Nickel	**7.5×10^{-4}**	**600**	**Ferromagnetic**
Cobalt	**3.1×10^{-4}**	**250**	**Ferromagnetic**
Water	1.26×10^{-6}	0.9999992	Diamagnetic
Wood	1.26×10^{-6}	1.00000043	Paramagnetic

Induced Magnetism - A ferromagnetic material with no net magnetism will, when placed in an external magnetic field, become magnetized. This induced magnetism is created when the atoms in the material align with the external field so that the induced magnetic field is in the same direction as the external field and adds to it. The induced field is created in ferromagnetic materials when the domains (small areas with the same magnetic field direction) are aligned by the external field.



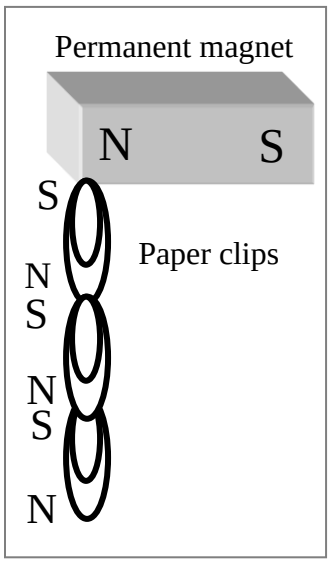
- This alignment adds to the external field creating a lager field.
- The domains can never be aligned in the opposite directions as the external field so there can never be a repulse induced force.
- When the Magnetic field is removed the atomic magnetic fields may remain aligned to produce a permanent magnet or go back to random domains (non-magnetic).

This induced field can be very large, much larger than the field that induced it.

- This is why ferromagnetic material will "stick" to other ferromagnetic material even if the second material is not magnetized. Magnet to refrigerator door.
- This also explains why magnets will not be attracted to non-ferromagnetic material like aluminum or copper because there is no induced field to create a force.
- Heating of a magnet randomizes the atoms thereby removing the magnetic properties of the material, so they are randomized and are no longer aligned, there is no longer a net field.

- Both ends of the magnet can induce a magnetic field in a ferromagnetic material, this is why it does not matter which end of the magnet is used to stick to the refrigerator.
- Induced magnetism has no repulsive force.

Example: If paper clips (iron) are placed on the North pole of a magnet then the magnet induces a magnetic field in the paper clip that puts the South pole with the North pole of the magnet and the north pole at the other end, the same this happens with the second and third paper clip. Paper clips that are not normally magnetic will stick together on a permanent magnet because of magnetic induction.



Making a magnet - By placing a ferromagnetic material in a strong magnetic field an induced magnetic can be produced but it does not remain very long after the magnetic field is removed. If you put the ferromagnetic material in a very strong magnetic field and then heat the material, the atoms will align with the external magnetic field, once the material is cooled off the aligned atoms stay longer because they are "frozen" in place.

Rare earth magnets such as Neodymium or Samarium-cobalt make very strong magnetics because their chemical structure allows the magnetic moments of the atoms to align, and rare earth elements can have up to 7 electrons with the same direction of spin in their outer orbits creating very large magnetic fields.

Bar Magnet – is any magnetic material that has a net magnetic field.

Ferromagnetism is a consequence of the magnetic properties of the electrons that all atoms contain. An electron behaves in some respects as through it is a spinning charged sphere, and it is therefore magnetically equivalent to a tiny current loop or bar magnet. In most substances the magnetic fields of the electrons cancel, but ferromagnetic substances the cancellation is not complete, and each atom has certain magnetic field of its own due to the electrons spin. The atoms magnetic fields can align themselves in groups called domains.



Not Magnetized

- In a material that is not magnetized, the magnetic domains point in random directions.

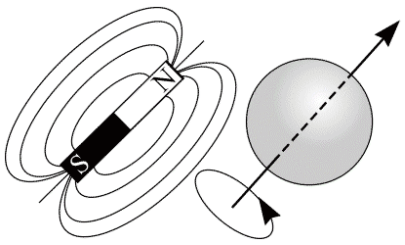


Magnetized

- In a magnetized material, all or most of the magnetic domains are arranged in the same direction.

Neodymium Magnets – Neodymium is a rare earth element that is used to make very strong permanent magnetics, the magnet is actually an alloy, using Neodymium, iron and boron. Because of the crystalline structure of the alloy the domains align easily along the crystal axis therefore create very strong magnetic fields and the domains are difficult to misalign, so the material stays magnetized.

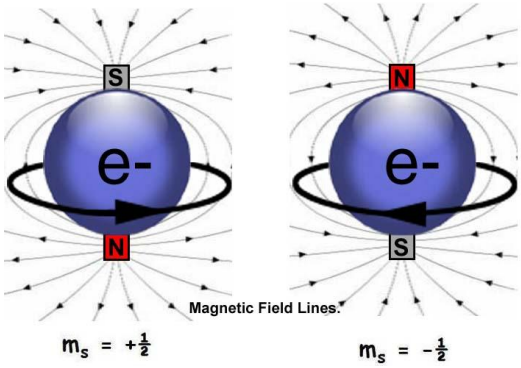
The electrons magnetic moment (or spin) - The electron generates both an electric field due to its charge, a magnetic field due to the electrons spinning around the nucleus of the atom (orbital magnetic moment) and the electron itself spinning like a top (electron spin). Both types of spinning create a magnetic field.



- Protons also create magnetic field, but the largest fields come from the electron because they are lighter, move around more and are on the outside of the atom.

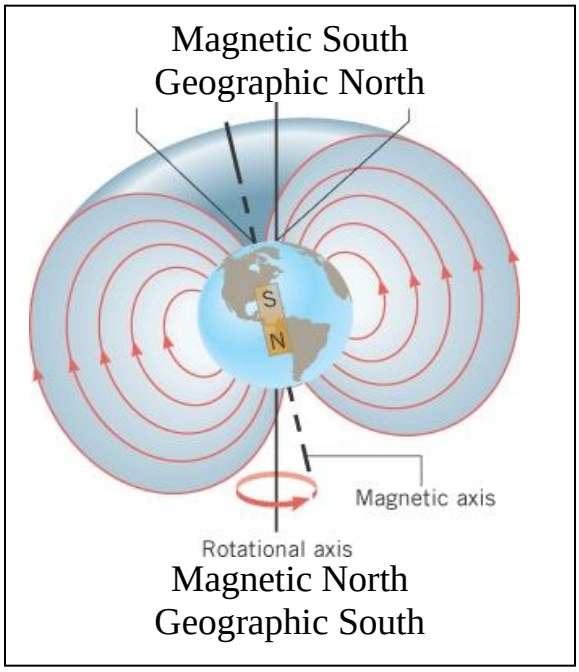
Electrons and protons can be thought of as subatomic magnets due to their spin.

The largest field is created by the electron itself spinning, this "spinning" creates the magnetic field because it is a moving charge. It acts much like a wire loop with a current (moving charge) in the shape of a circle, it creates what acts like a bar magnet with a north and South Pole. Depending on the direction the electron is spinning the south pole is either pointing up or down, for the south pole pointing up is “spin up” and for the south pole pointing down is called “spin down”. This is why for the orbits of the electrons around the atom, two electrons can occupy an energy level because the north pole lines up with the south pole of the other electron and create a stable orbit but if the north pole and north pole line up on the same side this would be unstable.



In most materials the electrons spin come in pairs, so the net magnetic field cancels out, but for some atoms, iron for instance, there are 4 electrons that have their fields in the same direction, so they add and give a net magnetic field. This net field is not very large for one atom but since there are billions and billions of atoms in any everyday substance the fields can become large. Not all iron forms magnets because these atoms are aligned only in small area or domains within the large block of iron and the net effect is no field or very little. But if the iron is placed in an external magnetic field, then these domains align and add to the eternal field creating very large magnetic fields.

Earth’s Magnetic field - The movement of the iron core of the earth acts as a kind of current loop that produces the magnetic field which is known as the Geomagnetic Dynamo at the outer core of the earth.



- Earth can be thought of as having a permanent magnet in its core with the north magnetic pole at the geographic South Pole and the Magnetic south pole at the geographic North Pole
- The needle of a compass on the surface of the earth aligns along the Magnetic field lines and the north pole of the compass magnet points in the direction of the south magnetic pole which is the north geographic pole.
- The Magnetic field is not horizontal to the earth's surface everywhere over the entire Earth but is some angle to the horizontal (60° below the horizontal for UNF)
- The Earth's Magnetic field is tilted about 11 degrees from the geographic poles and is changing.
- The Magnetic field strength of the earth is about 3×10^{-5} Tesla at sea level.
- The Magnetosphere: The magnetic field of the earth forms a protective "bubble" called the magnetosphere, which deflects most of the solar wind, a stream of charged particles from the Sun. If there was not a magnetic field life on earth would be very different if at all.
- The earth's magnetic field has switch ≈ 100 times in last 5 million years. The magnetic field of the earth last flipped 780,000 years ago.
- The magnetic south pole (geographic North) is moving from northern Canada towards Siberia at a rate of approximately 30 miles per year.
- Many animals, including migratory birds, sea turtles, and certain bacteria, use the field for magnetoreception to navigate across the globe.

➢ Stars (the Sun) and other planets can have magnetic fields.